

Task 2 — n8n API Integration Workflow README

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1. Data Pipeline Visual Canvas Layout

- **Step 1: Schedule Trigger**
Automatically initializes the pipeline execution cycle every 1 hour.
- **Step 2: Fetch Raw Data (HTTP Request)**
Performs an asynchronous HTTP GET request to the target JSONPlaceholder API. It is configured with a Continue On Fail error mitigation boundary to ensure safety.
- **Step 3: Keep Top 5 Items (Code Node)**
Transforms and filters the incoming array payload using a clean JavaScript data slice operation to keep only the top 5 records.
- **Step 4: Threshold Split (IF Conditional Router)**
Tracks condition rules based on the primary item ID to dynamically branch data. Items matching the numeric limit route to the True Vector Branch, while remaining records route to the False Vector Branch.

2. Integration Details and Pipeline Logic

- **APIs Used:** This data automation framework integrates the open-source JSONPlaceholder REST API layer. This target was chosen due to its high availability and clean nested array data structures, which match production-grade e-commerce payloads.
- **Transformation Layer:** The data processing chunk links a pure JavaScript execution runtime block. It runs a clean array slice operation to truncate the incoming payload matrix. This filters out database noise and keeps only the top 5 trending content records.
- **Conditional Threshold Routing:** The pipeline branches data features dynamically using an automated binary comparison operation. If a processed data item structural ID exceeds the numeric threshold limit, it maps down the True vector path. Otherwise, it diverts to the fallback lane.

3. Production Error Mitigation

To satisfy the constraint that the workflow must never crash silently, the critical Data Extraction HTTP Request layer is hardcoded with strict Continue Regular Output error

intercept boundaries. If an unexpected runtime exception, network dropout, or endpoint timeout occurs, the system wrapper intercepts the error payload, flags the diagnostic exception code, and allows the remaining application threads to execute smoothly without crashing the core operational platform.